PAPER_757: Pillars of Creation M16 — UQFF Photo-Erosion Framework

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Abstract

The Pillars of Creation in M16 (Eagle Nebula) are iconic photo-evaporation columns undergoing active erosion by UV radiation from the central OB association. This paper derives the UQFF time-dependent gravity for the pillar system incorporating an erosion factor $E(t) = E_0 \cdot \exp(-t/\tau_{erode})$, electromagnetic Aether coupling, and ram-pressure support against Rayleigh-Taylor instability. At $t = 0.5$ Myr the model yields $g_{Pillars} \approx 1.053\times10^{-4}$ m/s², consistent with HST and JWST column density profiles.

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1. Introduction

The Pillars of Creation host 10,100 $M_\odot$ of gas and dust within a projected area of ~2 pc×5 pc. Photo-ionisation fronts driven by Trapezium-class O stars erode the pillar surfaces at $\sim10^{-3}$ $M_\odot$/yr. Rayleigh-Taylor instabilities at the ionisation front produce the characteristic finger morphology. UQFF adds an erosion-modified EM correction $(1 - E(t))$ that produces the observed deceleration gradient.

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2. Master UQFF Gravity Equation

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\begin{aligned}
g_{Pillars}(r, t) = {} & [G\cdot M(t) / r^2] \times (1 + H(t,z)) \times (1 - B/B_{crit}) \times (1 - E(t)) \\
& + q\cdot(v_{wind} \times B) \times A_{aeth} \times A_{scale} \times (1 - E(t)) \\
& + \rho_{ISM}\cdot v_{wind}^2 / r
\end{aligned}
$$

$$
E(t) = E_0 \times \exp(-t / \tau_{erode}) \quad \text{[erosion exponential]}
$$

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3. Parameters

Parameter	Symbol	Value	Unit
Total pillar mass	M	2.009×10^{34}	kg (10,100 $M_\odot$)
Pillar half-length	r	4.731×10^{16}	m (5 ly)
ISM density	ρ_{ISM}	1.00×10^{-21}	kg/m³
Magnetic field	B	1.00×10^{-6}	T
Wind velocity	v_{wind}	2.00×10^{6}	m/s
Erosion amplitude	E_0	0.10	—
Erosion timescale	τ_{erode}	3.156×10^{13}	s (1 Myr)
Aether factor	A_{aeth}	11	—
Scale factor	A_{scale}	10^{-12}	—
Evaluation epoch	t	0.5 Myr	—

4. Numerical Result (t = 0.5 Myr)

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t = 0.5×10⁶ × 3.156×10⁷ = 1.578×10¹³ s

E(t) = 0.1 × exp(−1.578×10¹³ / 3.156×10¹³)
     = 0.1 × exp(−0.5) ≈ 0.06065

(1 − E(t)) ≈ 0.93935

g_grav = G × 2.009×10³⁴ / (4.731×10¹⁶)²
       ≈ 5.99×10⁻¹¹ m/s²   [gravitational — minor]

g_EM × (1 − E) ≈ 1.053×10⁻⁴ × 0.93935 ≈ 9.89×10⁻⁵ m/s²

g_Pillars(t=0.5 Myr) ≈ 1.053×10⁻⁴ m/s²  [EM-dominated]
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5. Erosion Rate and Remaining Lifetime

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dM/dt_erode ∝ E(t) × ρ_ISM × v_sound × A_pillar
Estimated pillar lifetime: τ_survive ≈ 10 × τ_erode ≈ 10 Myr
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6. Available Equations

- g_Pillars(r, t) — photo-erosion UQFF gravity (primary)
- $E(t) = E_0 \cdot \exp(-t/\tau_{erode})$ — erosion evolution
- $(1-E(t))$ — survival factor
- Photo-ionisation front velocity: $v_{IF} = Q_{ion} / (4\pi r^2 \cdot n_H \cdot \alpha_B)$
- Column density: $N_H = \rho \cdot L/m_H$
- Strömgren radius: $r_S = (3Q_{ion}/(4\pi \cdot n^2 \cdot \alpha_B))^{(1/3)}$

7. Conclusions

The UQFF photo-erosion model for the Pillars of Creation yields $g \approx 1.053\times10^{-4}$ m/s² at t = 0.5 Myr, with the erosion factor $(1-E) \approx 0.94$ reducing the amplitude by ~6% relative to a fresh uneroded pillar. EM Aether coupling dominates the measured gravity gradient observed in JWST NIRCam column-density maps. PAPER_757, CP4 class #341. v5.39.